

5 Converting the Image Format

5.1 Converting the Image Format Using qemu-img

Scenarios

You can import an image file in VHD, VMDK, QCOW2, RAW, VHDX, QCOW, VDI, QED, ZVHD, or ZVHD2 format to HUAWEI CLOUD. Image files in other formats need to be converted before being imported. The open-source tool **qemu-img** is provided for you to convert image file formats.

Background

- **qemu-img** supports the mutual conversion of image formats VHD, VMDK, QCOW2, RAW, VHDX, QCOW, VDI, and QED.
- ZVHD and ZVHD2 are self-developed image file formats and cannot be identified by **qemu-img**. To convert image files to any of the two formats, use the **qemu-img-hw** tool. For details, see [Converting the Image Format Using qemu-img-hw](#)
- When you run the command to convert the format of VHD image files, use VPC to replace VHD. Otherwise, qemu-img cannot identify the image format. For example, to convert a CentOS 6.9 VHD image file into a QCOW2 image file, run the following command:
qemu-img convert -p -f vpc -O qcow2 centos6.9.vhd centos6.9.qcow2

Windows

1. Install qemu-img.
 - a. Download the qemu-img installation package from <https://qemu.weilnetz.de/w64/>.
 - b. Double-click the setup file to install qemu-img in **D:\Program Files\qemu** (an example installation path).
2. Configure environment variables.
 - a. Choose **Start > Computer** and right-click **Properties**.

- b. Click **Advanced system settings**.
- c. In the **System Properties** dialog box, click **Advanced > Environment Variables**.
- d. In the **Environment Variables** dialog box, search for **Path** in the **System Variable** area and click **Edit**. Add **D:\Program Files\qemu** to **Variable Value**. Use semicolons (;) to separate variable values.

 NOTE

If **Path** does not exist, add it and set its value to **D:\Program Files\qemu**.

- e. Click **OK**.
3. Verify the installation.
Choose **Start > Run**, enter **cmd**, and press **Enter**. In the **cmd** window, enter **qemu-img --help**. If the qemu-img version information is contained in the command output, the installation is successful.
4. Convert the image format.

- a. In the **cmd** window, run the following commands to switch to **D:\Program Files\qemu**:
d:
cd D:\Program Files\qemu
- b. Run the following command to convert the image file format from VMDK to QCOW2:

```
qemu-img convert -p -f vmdk -O qcow2 centos6.9.vmdk centos6.9.qcow2
```

The parameters are described as follows:

- **-p** indicates the image conversion progress.
- **-f** indicates the source image format.
- The part following **-O** (which must be in upper case) consists of the required format, source image file, and target image file.

After the conversion is complete, the target image file is displayed in the directory where the source image file is located.

The following information is displayed:

```
# qemu-img convert -p -f vmdk -O qcow2 centos6.9.vmdk centos6.9.qcow2
(100.00/100%)
```

- c. Run the following command to query details about the converted image file in QCOW2 format:

```
qemu-img info centos6.9.qcow2
```

The following information is displayed:

```
# qemu-img info centos6.9.qcow2
image: centos6.9.qcow2
file format: qcow2
virtual size: 1.0G (1073741824 bytes)
disk size: 200K
cluster_size: 65536
Format specific information:
  compat: 1.1
  lazy refcounts: false
```

Linux

1. Install qemu-img.
 - For Ubuntu or Debian, run the following command:
apt install qemu-img
 - For CentOS, Red Hat, or Oracle, run the following command:
yum install qemu-img
 - For SUSE or openSUSE, run the following command:
zypper install qemu-img
2. Run the following command to check whether the installation is successful:
qemu-img -v

If the version information and help manual of the qemu-img tool are contained in the command output, the installation is successful. If CentOS 7 is used, the command output is as follows:

```
[root@CentOS7 ~]# qemu-img -v
qemu-img version 1.5.3, Copyright (c) 2004-2008 Fabrice Bellard
usage: qemu-img command [command options]
QEMU disk image utility

Command syntax:
check [-q] [-f fmt] [--output=ofmt] [-r [leaks | all]] [-T src_cache] filename
create [-q] [-f fmt] [-o options] filename [size]
commit [-q] [-f fmt] [-t cache] filename
compare [-f fmt] [-F fmt] [-T src_cach]
```

3. Convert the image format. For example, perform the following steps to convert a VMDK image file running CentOS 7 to a QCOW2 image file:
 - a. Run the following command to convert the image file format to QCOW2:
qemu-img convert -p -f vmdk -O qcow2 centos6.9.vmdk centos6.9.qcow2

The parameters are described as follows:

- **-p**: indicates the conversion progress.
- **-f** indicates the source image format.
- The pat following **-O** (which must be in upper case) is the converted image format + source image file name + target image file name.

After the conversion is complete, the target image file is displayed in the directory where the source image file is located.

The following information is displayed:

```
[root@CentOS7 home]# qemu-img convert -p -f vmdk -O qcow2 centos6.9.vmdk
centos6.9.qcow2
(100.00/100%)
```

- b. Run the following command to query details about the converted image file in QCOW2 format:
qemu-img info centos6.9.qcow2

The following information is displayed:

```
[root@CentOS7 home]# qemu-img info centos6.9.qcow2
image: centos6.9.qcow2
file format: qcow2
virtual size: 1.0G (1073741824 bytes)
disk size: 200K
```

```
cluster_size: 65536
Format specific information:
  compat: 1.1
  lazy refcounts: false
```

Examples

A pre-allocated image depends on two files: **xxxx.vmdk** (configuration file) and **xxxx-flat.vmdk** (data file) and cannot be directly imported to the cloud platform. When you export a pre-allocated image file in VMDK monolithic Flat format from the VMware platform, you must convert its format to common VMDK or QCOW2 before it can be imported to the cloud platform.

The following uses the image files **centos6.9-64bit-flat.vmdk** and **centos6.9-64bit.vmdk** as an example to describe how to use **qemu-img** to convert image formats.

1. Run the following commands to query the image file details:

```
ls -lh centos6.9-64bit*
```

```
qemu-img info centos6.9-64bit.vmdk
```

```
qemu-img info centos6.9-64bit-flat.vmdk
```

The following information is displayed:

```
[root@CentOS7 tmp]# ls -lh centos6.9-64bit*
-rw-r--r--. 1 root root 10G Jun 13 05:30 centos6.9-64bit-flat.vmdk
-rw-r--r--. 1 root root 327 Jun 13 05:30 centos6.9-64bit.vmdk
[root@CentOS7 tmp]# qemu-img info centos6.9-64bit.vmdk
image: centos6.9-64bit.vmdk
file format: vmdk
virtual size: 10G (10737418240 bytes)
disk size: 4.0K
Format specific information:
  cid: 3302005459
  parent cid: 4294967295
  create type: monolithicFlat
  extents:
    [0]:
      virtual size: 10737418240
      filename: centos6.9-64bit-flat.vmdk
      format: FLAT
[root@CentOS7 tmp]# qemu-img info centos6.9-64bit-flat.vmdk
image: centos6.9-64bit-flat.vmdk
file format: raw
virtual size: 10G (10737418240 bytes)
disk size: 0
```

NOTE

The command output shows that the format of **centos6.9-64bit.vmdk** is VMDK and that of **centos6.9-64bit-flat.vmdk** is RAW. You can convert the format of only **centos6.9-64bit.vmdk**. For details about how to convert it, see [3](#).

2. Run the following command to query the configuration of the pre-allocated image file:

```
cat centos6.9-64bit.vmdk
```

The following information is displayed:

```
[root@CentOS7 tmp]# cat centos6.9-64bit.vmdk
# Disk DescriptorFile
version=1
CID=c4d09ad3
parentCID=ffffff
createType="monolithicFlat"
```

```
# Extent description
RW 20971520 FLAT "centos6.9-64bit-flat.vmdk" 0
```

```
# The Disk Data Base
#DDB
```

```
ddb.virtualHWVersion = "4"
ddb.geometry.cylinders = "20805"
ddb.geometry.heads = "16"
ddb.geometry.sectors = "63"
ddb.adapterType = "ide"
```

3. Place **centos6.9-64bit-flat.vmdk** and **centos6.9-64bit.vmdk** in the same directory. Run the following command to convert the format of **centos6.9-64bit.vmdk** to QCOW2 using **qemu-img**:

```
[root@CentOS7 tmp]# qemu-img convert -p -f vmdk -O qcow2 centos6.9-64bit.vmdk
centos6.9-64bit.qcow2
(100.00/100%)
```

4. Run the following command to query details about the converted image file in QCOW2 format:

```
qemu-img info centos6.9-64bit.qcow2
```

The following information is displayed:

```
[root@CentOS7 tmp]# qemu-img info centos6.9-64bit.qcow2
image: centos6.9-64bit.qcow2
file format: qcow2
virtual size: 10G (10737418240 bytes)
disk size: 200K
cluster_size: 65536
Format specific information:
  compat: 1.1
  lazy refcounts: false
```

Follow-up Procedure

After the image file format is converted, you can use **obsutil** to upload the image file to an OBS bucket for creating a private image. For details about how to use **obsutil** to upload an image file, see [Uploading an Object](#).

5.2 Converting the Image Format Using **qemu-img-hw**

Scenarios

You can import an image file in VHD, VMDK, QCOW2, RAW, VHDX, QCOW, VDI, QED, ZVHD, or ZVHD2 format to HUAWEI CLOUD. Image files in other formats need to be converted into any of these formats using the open-source tool **qemu-img** before being imported. However, the **qemu-img** tool cannot convert image files to the ZVHD or ZVHD2 format. To convert image files to any of the two formats, use the self-developed tool **qemu-img-hw**. This section describes how to use **qemu-img-hw** to convert an image file to ZVHD2.

Background

qemu-img-hw can be used only in Linux. You can run it on a local Linux server or a Linux ECS on the cloud platform. The following procedure uses an EulerOS ECS as an example.

Procedure

1. Upload the image file to be converted to the ECS.
 - If the local host runs a Linux OS, run the **scp** command.
For example, to upload **image01.qcow2** to the **/usr/** directory on the ECS, run the following command:
scp /var/image01.qcow2 root@xxx.xxx.xx.xxx:/usr/
xxx.xxx.xx.xxx indicates the EIP bound to the ECS.
 - If the local host runs a Windows OS, use a file transfer tool, such as WinSCP, to upload the image file to the ECS.
2. Obtain the **qemu-img-hw** software package, upload it to the ECS, and then decompress the package.

Table 5-1 qemu-img-hw package

Tool Package	How to Obtain
qemu-img-hw.zip	https://cn-south-1-cloud-reset-pwd.obs.cn-south-1.myhuaweicloud.com/imageImportTools/qemu-img-hw.zip

NOTE

This tool can be used only on x86 servers.

3. Convert the image format.
 - a. Go to the directory where **qemu-img-hw** is stored, for example, **/usr/qemu-img-hw**.
cd /usr/qemu-img-hw
 - b. Run the following command to change file permissions:
chmod +x qemu-img-hw
 - c. Run the **qemu-img-hw** command to convert the image file to the ZVHD2 format.
The command format of **qemu-img-hw** is as follows:
./qemu-img-hw convert -p -O Target_image_format Source_image_file Target_image_file
For example, run the following command to convert an **image01.qcow2** file to an **image01.zvhd2** file:
./qemu-img-hw convert -p -O zvhd2 image01.qcow2 image01.zvhd2

Appendix 1: Common qemu-img-hw Commands

- Converting image file formats: **qemu-img-hw convert -p -O Target_image_format Source_image_file Target_image_file**
The parameters are described as follows:
-p: indicates the conversion progress.

The part following **-O** (which must be in upper case) consists of the target image format, source image file, and target image file.

For example, run the following command to convert a QCOW2 image file to a ZVHD2 file:

```
qemu-img-hw convert -p -O zvhd2 test.qcow2 test.zvhd2
```

- Querying image file information: **qemu-img-hw info *Image file***
An example command is **qemu-img-hw info test.zvhd2**.
- Viewing help information: **qemu-img-hw -help**

Appendix 2: Common Errors During qemu-img-hw Running

- Symptom:

The following information is displayed when you run the **qemu-img-hw** command:

```
./qemu-img-hw: /lib64/libc.so.6: version `GLIBC_2.14' not found (required by ./qemu-img-hw)
```

Solution:

Run the **strings /lib64/libc.so.6 | grep glibc** command to check the glibc version. If the version is too early, install the latest version. Run the following commands in sequence:

```
wget http://ftp.gnu.org/gnu/glibc/glibc-2.15.tar.gz
```

```
wget http://ftp.gnu.org/gnu/glibc/glibc-ports-2.15.tar.gz
```

```
tar -xvf glibc-2.15.tar.gz
```

```
tar -xvf glibc-ports-2.15.tar.gz
```

```
mv glibc-ports-2.15 glibc-2.15/ports
```

```
mkdir glibc-build-2.15
```

```
cd glibc-build-2.15
```

```
../glibc-2.15/configure --prefix=/usr --disable-profile --enable-add-ons --with-headers=/usr/include --with-binutils=/usr/bin
```

NOTE

If **configure: error: no acceptable C compiler found in \$PATH** is displayed, run the **yum -y install gcc** command.

```
make
```

```
make install
```

- Symptom:

The following information is displayed when you run the **qemu-img-hw** command:

```
./qemu-img-hw: error while loading shared libraries: libaio.so.1: cannot open shared object file: No such file or directory
```

Solution: Run the **yum install libaio** command.

Follow-up Procedure

After the image file format is converted, you can use obsutil to upload the image file to an OBS bucket for creating a private image. For details about how to use obsutil to upload an image file, see [Uploading an Object](#).